

1. Contrapositive: if x is odd then 8 divides $x^2 - 1$

Suppose x is odd

So $x = 2k + 1$, k is any integer

$$\begin{aligned} \text{Then } x^2 - 1 &= (2k + 1)^2 - 1 \\ &= 4k^2 + 4k + 1 - 1 \\ &= 4k^2 + 4k \end{aligned}$$

$$\frac{x^2 - 1}{8} = \frac{4k^2 + 4k}{8} = \frac{k(k+1)}{2}, \text{ if } k \text{ is even or odd, the result would be even.}$$

which makes the whole thing ~~di~~ even and divisible by 8.

Hence, result follows.

2. Let our 4 digit number be ~~abba~~. x, y, y, x
We know that in a palindrome

$$x_1 = x_2 \text{ and } y_1 = y_2$$

We can write x, y, y, x as

$$= 1000x + 100y + 10y + x$$

Rearranging and substituting we get

$$A = x(1000 + 1) + y(100 + 10)$$

$$A = x, (1001) + y, (110)$$

$$A = x, (91 \times 11) + y, (11 \times 10)$$

$$A = 11 (x, (91) + y, (10))$$

If we divide A with 11 , we get ~~an~~ an answer.

Hence, it is proved.

$$3. (\exists! x \in \mathbb{R})(\forall y \in \mathbb{R})(y < x \Rightarrow (x+y < 0) \wedge (x-y > 0))$$

Assume that $x < 0$, we also know that $y \in \mathbb{R}$ and $y < x$.

Then it follows the statement $x-y > 0$ because it is always less than 0 , and x is less than 0 , the difference will always be greater than 0 .

For $x+y < 0$, since x and y both are negative this statement cannot be true.

$$4. (\forall x \in \mathbb{R})(\forall y \in \mathbb{R})(y < x \Rightarrow (\exists m \in \mathbb{R})(\forall n \in \mathbb{N}) \\ (n > m \Rightarrow \frac{1}{n} < x - y))$$

We have $y < x$, which makes $x = y + \epsilon$ for
some
 $\epsilon > 0$

Assume $m = \frac{1}{\epsilon}$, $m \in \mathbb{R}$ and $n \in \mathbb{N}$

it is given that $n > m$, $\epsilon = x - y$

$$n > m = \frac{1}{\epsilon}$$

$$\frac{1}{n} < \epsilon$$

$$\frac{1}{n} < (x - y)$$

This completes the proof